# ARCHIT BANKEY

MALE, 21/04/2006

## TECHNICAL COMPETENCIES

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| **Programming** | **Languages:** Python, SQL |
| **Machine Learning** | **Frameworks & Libraries:** Scikit-learn, XGBoost, PyTorch, Ultralytics YOLOv8 |
| **Data & Vision** | **Tools:** Pandas, NumPy, OpenCV, Roboflow, SHAP, Matplotlib, Seaborn |
| **Dev & Tools** | **Environment:** Jupyter Notebook, Google Colab, GitHub, Git, VS Code |
| **Generative AI** | **Skills:** Prompt Engineering, LLM integration concepts, IBM Generative AI stack |

## PROJECTS

**Social Media Addiction & Mental Health Classifier** *ML Pipeline | Classification*

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| **Overview** | Built an end-to-end **3-class mental health classification system** (Low / Moderate / High risk) to stratify student risk tiers from social media behavioural and demographic survey data. |
| **Workflow & Optimization** | · **Model Development:** Designed and trained Logistic Regression, Random Forest, and XGBoost classifiers on a real student survey dataset, achieving **90% test accuracy** and **88% cross-validated accuracy.**<br>· **Feature Engineering:** Engineered target variable from raw mental health scores; enforced **feature-leakage prevention** by identifying and removing 5 correlated downstream columns to ensure model integrity.<br>· **Evaluation Rigour:** Applied **5-fold Stratified Cross-Validation** with Macro F1 scoring to handle class imbalance; XGBoost selected as production model with low CV std of ±0.01, confirming generalisation stability. |
| **Outcome** | Proved that social media usage patterns are **4–5x more influential** than personal demographics in determining student mental health outcomes. |

**Hourly Energy Consumption Forecasting** *ML Pipeline | Time Series Regression*

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| **Overview** | Built an end-to-end ML pipeline on the PJM Interconnection dataset to predict 24-hour-ahead energy demand — the core problem utilities face in deciding how much power to generate. |
| **Workflow & Optimization** | · **Data Cleaning & EDA** Removed anomalous readings and uncovered cyclical patterns (hour, day, month) and a long-term declining trend — directly driving feature selection.<br>· **Feature Engineering**: Built cyclical encodings (hour_sin/hour_cos), an hour × month interaction term, and replaced raw lag with lag_24h_delta to **prevent the model from simply copying yesterday's values**..<br>· **Metric Integrity**: **Achieved >90% forecasting** accuracy ($R^2$ score) with **low RMSE and MAE** through feature engineering, cross-validation, and regression-based energy demand prediction. |
| **Outcome** | Delivered a demand prediction pipeline that enables informed **generation planning before the day begins** — reducing the risk of blackouts and costly overproduction. |

**Real-Time Weapon Detection System** *Computer Vision | Object Detection*

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| **Overview** | Fine-tuned **YOLOv8m** on a custom multi-class weapon dataset to build a production-grade real-time detection system for security surveillance applications. |
| **Workflow & Optimization** | · **Model Fine-Tuning:** Fine-tuned YOLOv8m on a custom Roboflow dataset (knife, gun, pistol classes) for 150 epochs ; trained on GPU at **640×640 resolution**, batch size 16.<br>· **Inference Pipeline:** Engineered **real-time inference pipeline** supporting static images, recorded video, and live webcam feed via OpenCV — with dynamic confidence threshold slider and automated alert-frame archiving on detection.<br>· **Evaluation:** Benchmarked model performance using **mAP50, mAP50-95, Precision, and Recall** via a dedicated validation script, adhering to standard object detection evaluation protocols. |
| **Outcome** | **Transforms reactive CCTV cameras into proactive security systems** capable of instantly alerting authorities the split-second a weapon is drawn, potentially stopping a tragedy before it happens. |

## ACADEMIC PROFILE

| Degree / Board | Institution | Score | Year |
|---|---|---|---|
| B.Tech. (AI/ML) | MIT Art, Design & Technology, Pune | 8.6 CGPA | 2024 – 28 |
| Class XII (CBSE) | Sardar Patel Public School , Bhopal | 74% | 2024 |

## CERTIFICATIONS & CREDENTIALS

| Issuing Body | Certification | Date |
|---|---|---|
| **IBM / Coursera** | **Generative AI**: Elevate Your Data Science Career \| *Verify:* *coursera.org/verify/XOSNYVBEA2OH* | Mar 2026 |
| **IBM / Coursera** | **Machine Learning with Python** \| *Verify: coursera.org/verifyDQLOZIFK17DG* | Jan 2026 |

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